Model Predictive Control

Chapter 7: Practical Issues

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Outline

1. MPC: Practical Issues

2. Reference Tracking

3. Enlarging the Feasible Set

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1. MPC: Practical Issues

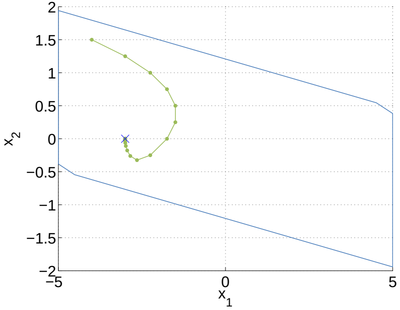
2. Reference Tracking

3. Enlarging the Feasible Set

MPC: Practical Issues

Tracking

- Classic MPC problem: Regulation to the origin
- Common task in practice: Tracking of non-zero output set points



Example: Double Integrator

$$x(k+1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x(k) + \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} u(k)$$

$$-0.5 \leq u \leq 0.5$$

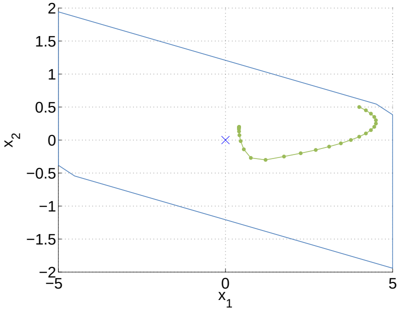
$$-5 \leq x_i \leq 5, \; i = 1, 2$$

$\rightarrow$ Want to use MPC for tracking

MPC: Practical Issues

Disturbance rejection

- Constant disturbance causes offset from the origin / the desired set point



Example: Double Integrator

$$x(k+1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x(k) + \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} u(k) + d$$

$d = 0.2$

$-0.5 \leq u \leq 0.5$

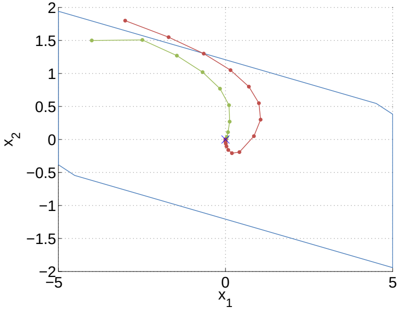
$-5 \leq x_i \leq 5,\ i = 1, 2$

$\rightarrow$ Want to remove offset such that system converges to desired set point.

MPC: Practical Issues

Feasible Set

- Constraints restrict the set of states for which the optimization problem is feasible.
- MPC controller is only defined in the feasible set, where a solution exists



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$$-0.5 \leq u \leq 0.5$$

$$-5 \leq x_i \leq 5, \ i = 1, 2$$

$\rightarrow$ Want the feasible set to be as large as possible

Learning Objectives

- Understand difference between MPC for tracking and regulation
- Formulate MPC tracking problem
- Address constant disturbances to achieve offset-free tracking/regulation
- Analyze MPC without terminal set
- Formulate soft-constrained MPC problem and understand its properties

Outline

Tracking Problem

Consider the linear system model

$$x(k+1) = Ax(k) + Bu(k)$$

where $x \in \mathbb{R}^{n_x}$, $u \in \mathbb{R}^{n_u}$.

- Simplifying assumption: State can be measured.
- Constraints: $x \in \mathcal{X}$, $u \in \mathcal{U}$ with constraint sets

$$\mathcal{X} = \{x \mid G_x x \leq h_x\}, \mathcal{U} = \{x \mid G_u u \leq h_u\}$$

Goal: Track given reference r such that $z(k) = Hx(k) \rightarrow r \in \mathbb{R}^{n_r}$ as $k \rightarrow \infty$.

Note: Using this framework we can track sequence of constant targets, i.e. a piecewise constant reference. We will not cover time-varying references in this class.

Tracking Problem: Introduction

Standard MPC problem regulates system state to the origin

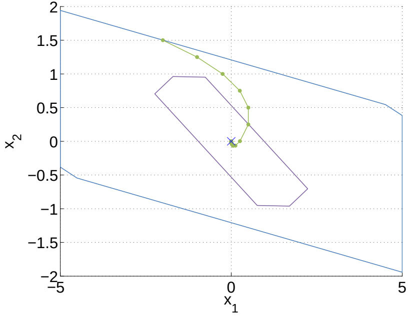
$$
\begin{aligned}
U^\star(x(k)) := \underset{U_k}{\operatorname{argmin}}\ & l_f(x_N) + \sum_{i=0}^{N-1} l(x_{k+i}, u_{k+i}) & \\
\text{subj. to}\ & x_k = x(k) & \text{measurement} \\
& x_{k+i+1} = Ax_{k+i} + Bu_{k+i} & \text{system model} \\
& x_{k+i} \in \mathcal{X} & \text{state constraints} \\
& u_{k+i} \in \mathcal{U} & \text{input constraints} \\
& U_k = \{u_k, u_{k+1}, \ldots, u_{k+N-1}\} & \text{optimization variables}
\end{aligned}
$$

Common task: Tracking of non-zero references

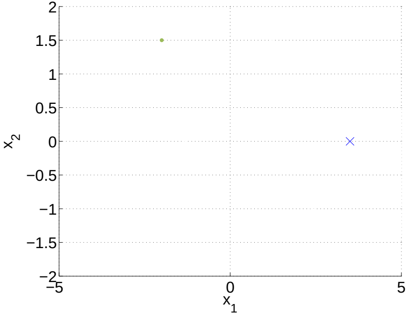
$\Rightarrow$ How can we modify the MPC problem to achieve tracking?

Tracking Problem: Introduction

Regulation to origin:



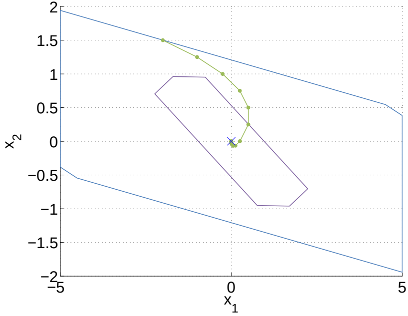
Non-zero target:



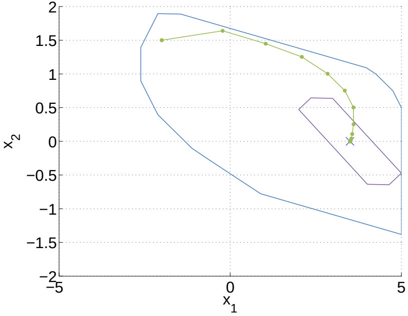
- Given target r, how do we obtain a corresponding state target x_s?

Tracking Problem: Introduction

Regulation to origin:



Non-zero target:



- Given target r, how do we obtain a corresponding state target x_s?
- How do we adapt MPC cost to control the system to the target state?
- How do we choose terminal set and when is MPC problem feasible with respect to target?

Outline

2. Reference Tracking

   The Steady-State Target Problem

   MPC for Reference Tracking

   MPC for Reference Tracking without Offset

Target State Corresponding to Output Reference

- The reference is achieved by the target state x_s if $z_s = Hx_s = r$
- Target state should be a steady-state, such that there exists an input that keeps system at target, i.e. $x_s = Ax_s + Bu_s$

$\Rightarrow$ Target condition:

$$\begin{aligned} x_s &= Ax_s + Bu_s \\ Hx_s &= r \end{aligned} \qquad \Longleftrightarrow \qquad \underbrace{\begin{bmatrix} I - A & -B \\ H & 0 \end{bmatrix}}_{(n_x+n_r)\times(n_x+n_u)} \begin{bmatrix} x_s \\ u_s \end{bmatrix} = \begin{bmatrix} 0 \\ r \end{bmatrix}$$

Steady-state Target Problem

- In the presence of constraints: (x_s, u_s) has to satisfy state and input constraints.
- In case of multiple feasible u_s, compute 'cheapest' steady-state (x_s, u_s) corresponding to reference r:

$$\begin{aligned} &\min\ u_s^T R_s u_s \\ &\text{subj. to } \begin{bmatrix} I-A & -B \\ H & 0 \end{bmatrix} \begin{bmatrix} x_s \\ u_s \end{bmatrix} = \begin{bmatrix} 0 \\ r \end{bmatrix} \\ &x_s \in \mathcal{X}, \quad u_s \in \mathcal{U}. \end{aligned}$$

- In general, we assume that the target problem is feasible
- If no solution exists: compute reachable set point that is 'closest' to r:

$$\begin{aligned} &\min\ (Hx_s - r)^T Q_s (Hx_s - r) \\ &\text{subj. to } x_s = Ax_s + Bu_s \\ &x_s \in \mathcal{X}, \quad u_s \in \mathcal{U}. \end{aligned}$$

Outline

2. Reference Tracking

   The Steady-State Target Problem

   MPC for Reference Tracking

   MPC for Reference Tracking without Offset

MPC for Reference Tracking

We now use control (MPC) to bring the system to a desired steady-state condition (x_s, u_s) yielding the desired output $z(k) \to r$.

The MPC is designed as follows

$$\min_{U} \|z_N - Hx_s\|^2_{P_z} + \sum_{i=0}^{N-1} \|z_i - Hx_s\|^2_{Q_z} + \|u_i - u_s\|^2_R$$

$$\text{subj. to [model, constraints]}$$

$$x_0 = x(k)$$

Delta-Formulation for Tracking

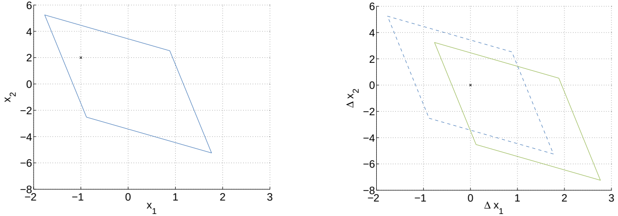
Idea: Treat set point tracking as regulation problem with a coordinate transformation

- Define deviation variables that (in the linear case) satisfy the same model equations:

$$\begin{aligned} \Delta x &= x - x_s \\ \Delta u &= u - u_s \end{aligned} \quad \Rightarrow \quad \begin{aligned} \Delta x_{k+1} &= x_{k+1} - x_s \\ &= Ax_k + Bu_k - (Ax_s + Bu_s) \\ &= A\Delta x_k + B\Delta u_k \end{aligned}$$

- Constraints for deviation variables:

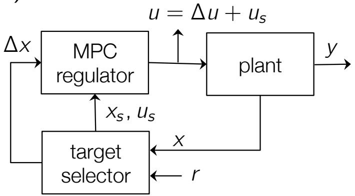
$$\begin{aligned} G_x x \le h_x &\Rightarrow \quad G_x \Delta x \le h_x - G_x x_s \\ G_u u \le h_u &\Rightarrow \quad G_u \Delta u \le h_u - G_u u_s \end{aligned}$$



MPC Problem for Tracking in Delta Formulation

- Obtain target steady-state corresponding to reference r. [1]
- Initial state $\Delta x(k) = x(k) - x_s$.
- Apply regulation problem to new system in Delta-Formulation:

$$
\begin{aligned}
\min \ & \sum_{i=0}^{N-1} \Delta x_i^T Q \Delta x_i + \Delta u_i^T R \Delta u_i + V_f(\Delta x_N) \\
\text{s.t. } & \Delta x_0 = \Delta x(k) \\
& \Delta x_{i+1} = A \Delta x_i + B \Delta u_i \\
& G_x \Delta x_i \leq h_x - G_x x_s \\
& G_u \Delta u_i \leq h_u - G_u u_s \\
& \Delta x_N \in \mathcal{X}_f
\end{aligned}
$$



- Find optimal sequence of ΔU^*
- Input applied to the system is $u_0^* = \Delta u_0^* + u_s$

[1] If the target steady-state is uniquely defined by the reference, we can also include the target condition as a constraint in the MPC problem.

MPC for Tracking: Convergence

Convergence

Assume target is feasible with $x_s \in \mathcal{X}, u_s \in \mathcal{U}$ and choose terminal weight $V_f(x)$ and constraint $\mathcal{X}_f$ as in the regulation case satisfying:

- $(A + BK)x \in \mathcal{X}_f$ for all $x \in \mathcal{X}_f$
- $V_f(x(k+1)) - V_f(x(k)) \leq -l(x(k), Kx(k))$ for all $x \in \mathcal{X}_f$

If in addition the target reference x_s, u_s is such that

- $x_s \oplus \mathcal{X}_f \subseteq \mathcal{X}$, $K\Delta x + u_s \in \mathcal{U}$ for all $\Delta x \in \mathcal{X}_f$

then the closed-loop system converges to the target reference,
i.e. $x(k) \to x_s$ and therefore $z(k) = Hx(k) \to r$ for $k \to \infty$

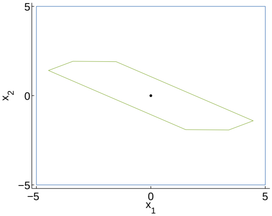
Proof: Choose local control law $\Delta u = K\Delta x$

- Invariance under local control law is directly inherited from regulation case
- Constraint satisfaction is provided by extra conditions:
  - $x_s \oplus \mathcal{X}_f \subseteq \mathcal{X} \;\rightarrow\; x \in \mathcal{X} \;\forall \Delta x = x - x_s \in \mathcal{X}_f$
  - $K\Delta x + u_s \in \mathcal{U} \;\forall \Delta x \in \mathcal{X}_f \;\rightarrow\; u \in \mathcal{U}$
- From asymptotic stability of the regulation problem: $\Delta x(k) \rightarrow 0$ for $k \rightarrow \infty$

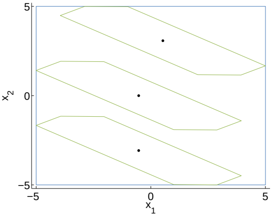
MPC for Tracking: Terminal Set

For the following consideration, consider only state constraints.

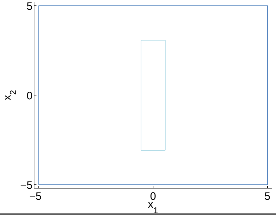
Regulation case:



Tracking using a shifted terminal set:



Set of feasible targets:

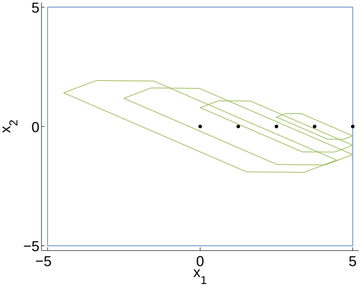


- Blue: State constraints
- Green: Terminal set

$\Rightarrow$ Set of feasible targets may be significantly reduced

MPC for Tracking: Terminal Set

Enlarge set of feasible targets by scaling terminal set for regulation



- Scale terminal set by scaling factor α, i.e. $\mathcal{X}_f^{\text{scaled}} = \alpha\mathcal{X}_f$
- Invariance is maintained: If $\mathcal{X}_f$ is invariant, then also $\alpha\mathcal{X}_f$
- Choose scaling factor α such that state and input constraints are still satisfied

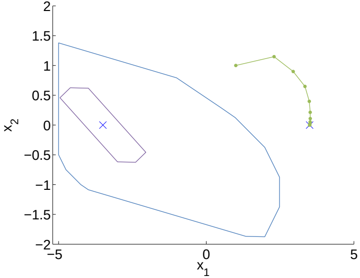
$\rightarrow$ Scaling is dependent on target

$\rightarrow$ All targets $x_s, u_s \in \mathcal{X} \times \mathcal{U}$ are feasible for constraints
Note: steady-state condition still limits the set of admissible targets

$\rightarrow$ For targets at the boundary of the constraints: $x_N = x_s$, which corresponds to a zero terminal set in the regulation case

Summary: MPC for Tracking

- Set point tracking problem corresponds to regulation problem after a coordinate transformation
- If the closed-loop system is stable, set point is achieved
- Stability can be guaranteed using the same tools as for regulation
- Difficulties:
  - Terminal set is a function of the target
  - Reference change can render the optimization problem infeasible



→ Recent approach for tracking resolves this issue
[MPC for tracking piecewise constant references for constrained linear systems, Limon et al., Automatica 2010]

  - Controller will show **offset** in case of unknown model error or disturbances.

Remarks: MPC for Tracking

- So far we have assumed that we can measure each state directly and the goal is $z(k) = Hx(k) \rightarrow r$ for $k \rightarrow \infty$.
- In practice, the state can not always be measured directly:

$$y(k) = Cx(k)$$

  i.e. the measured output $y(k) \in \mathbb{R}^{n_y}$ is a linear combination of the states.
- This is typically handled by designing a state estimator and running the MPC based on the estimated states.
- This approach is also used in offset-free tracking discussed in the following, where we assume the controlled variables to be a linear combination of measured outputs

$$z(k) = Hy(k) \rightarrow r \text{ for } k \rightarrow \infty.$$

Outline

2. Reference Tracking

   The Steady-State Target Problem

   MPC for Reference Tracking

   MPC for Reference Tracking without Offset

Constant disturbances

Constant disturbance is acting on the system causing system trajectory to deviate from nominal dynamics.

$$
\begin{aligned}
x(k+1) &= Ax(k) + Bu(k) + B_d d \\
y(k) &= Cx(k) + C_d d
\end{aligned}
$$

Objective: If system is stabilized in the presence of the disturbance then it converges to set point with zero offset.

Recall:

- In classic unconstrained control introduction of an integrating mode to remove offset
- Input constraints and actuator limitations require anti-windup techniques

Approach in constrained control:

- Model the disturbance
- Use the measurements and model to estimate the state and disturbance
- Find control inputs that use the disturbance estimate to remove offset

Augmented model

Incorporate disturbance model assuming integral disturbance dynamics

$$
\begin{aligned}
x_{k+1} &= Ax_k + Bu_k + B_d d_k \\
d_{k+1} &= d_k \\
y_k &= Cx_k + C_d d_k
\end{aligned}
$$

with $d \in \mathbb{R}^{n_d}$.

Only restriction on choice of B_d, C_d: observability of the augmented model

The augmented system is observable if and only if (A, C) is observable and

$$\begin{bmatrix} A - I & B_d \\ C & C_d \end{bmatrix} \text{ has full column rank, i.e. } \textbf{rank} = n_x + n_d$$

$\Rightarrow$ Maximal dimension of the disturbance: $n_d \leq n_y$

Intuition: At steady-state $\begin{bmatrix} A - I & B_d \\ C & C_d \end{bmatrix} \begin{bmatrix} x_s \\ d_s \end{bmatrix} = \begin{bmatrix} 0 \\ y_s \end{bmatrix}$ and given y_s, d_s must be uniquely defined.

Linear State Estimation

State observer for augmented model

$$\begin{bmatrix} \hat{x}(k+1) \\ \hat{d}(k+1) \end{bmatrix} = \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{x}(k) \\ \hat{d}(k) \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u(k) + \begin{bmatrix} L_x \\ L_d \end{bmatrix} (-y(k) + C\hat{x}(k) + C_d \hat{d}(k))$$

where $\hat{x}, \hat{d}$ are estimates of the state and disturbance, y the measured output.

Error dynamics:

$$\begin{aligned} \begin{bmatrix} x(k+1) - \hat{x}(k+1) \\ d(k+1) - \hat{d}(k+1) \end{bmatrix} &= \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix} \begin{bmatrix} x(k) - \hat{x}(k) \\ d(k) - \hat{d}(k) \end{bmatrix} \\ &\quad - \begin{bmatrix} L_x \\ L_d \end{bmatrix} (C\hat{x}(k) + C_d \hat{d}(k) - Cx(k) - C_d d(k)) \\ &= \left(\begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix} + \begin{bmatrix} L_x \\ L_d \end{bmatrix} \begin{bmatrix} C & C_d \end{bmatrix} \right) \begin{bmatrix} x(k) - \hat{x}(k) \\ d(k) - \hat{d}(k) \end{bmatrix} \end{aligned}$$

$\Rightarrow$ Choose $L = \begin{bmatrix} L_x \\ L_d \end{bmatrix}$ such that the error dynamics are stable and converge to zero, i.e. the estimator is asymptotically stable.

Offset-free Tracking

Lemma

Suppose the observer is asymp. stable and the number of outputs n_y equals the dimension of the constant disturbance n_d. The observer steady state satisfies:

$$\begin{bmatrix} A - I & B \\ C & 0 \end{bmatrix} \begin{bmatrix} \hat{x}_\infty \\ u_\infty \end{bmatrix} = \begin{bmatrix} -B_d \hat{d}_\infty \\ y_\infty - C_d \hat{d}_\infty \end{bmatrix}$$

where y_∞ and u_∞ are the steady state measured outputs and inputs.

$\Rightarrow$ Observer output $C\hat{x}_\infty + C_d \hat{d}_\infty$ tracks the measurement y_∞ without offset.

Offset-free Tracking

Goal: Track constant reference r, i.e. $z(k) = Hy(k) \to r$ for $k \to \infty$.

- New condition at steady-state:

$$x_s = Ax_s + Bu_s + B_d \hat{d}_\infty$$
$$z_s = H(Cx_s + C_d \hat{d}_\infty) = r$$

  - The system steady state is modified to account for effect of disturbance on state evolution
  - Target is modified to account for effect of disturbance on tracked variables
- Best forecast for steady-state disturbance is current estimate $\hat{d}_\infty = \hat{d}$
- Adapt target condition accordingly to account for disturbance:

$$\begin{bmatrix} A - I & B \\ HC & 0 \end{bmatrix} \begin{bmatrix} x_s \\ u_s \end{bmatrix} = \begin{bmatrix} -B_d \hat{d} \\ r - HC_d \hat{d} \end{bmatrix}$$

Note: Same procedure for the regulation case with $r = 0$

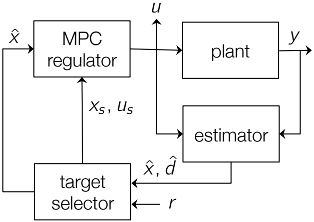
Offset-free Tracking

At each sampling time

1. Estimate state and disturbance $\hat{x}$, $\hat{d}$
2. Obtain (x_s, u_s) from steady-state target problem using disturbance estimate
3. Solve MPC problem for tracking using disturbance estimate $\hat{d}$:

$$\min_U \sum_{i=0}^{N-1} (x_i - x_s)^T Q (x_i - x_s) + (u_i - u_s)^T R (u_i - u_s) + V_f(x_N - x_s)$$

$$\begin{aligned}
\text{subj. to } & x_0 = \hat{x}(k) \\
& d_0 = \hat{d}(k) \\
& x_{i+1} = Ax_i + Bu_i + B_d d_i, \quad && i = 0, \dots, N \\
& d_{i+1} = d_i, && i = 0, \dots, N \\
& x_i \in \mathcal{X},\ u_i \in \mathcal{U}, && i = 0, \dots, N-1 \\
& x_N - x_s \in \mathcal{X}_f
\end{aligned}$$



$\rightarrow$ Can be rewritten in delta-formulation

Offset-free Tracking: Main Result

Denote by $\kappa(\hat{x}(k), \hat{d}(k), r) = u_0^\star$ the control law when the estimated state and disturbance are $\hat{x}(k)$ and $\hat{d}(k)$, respectively.

Theorem

Consider the case where the number of constant disturbances equals the number of outputs $n_d = n_y$. Assume the RHC is recursively feasible and unconstrained for $k \geq j$ with $j \in \mathbb{N}^+$ and the closed-loop system

$$
\begin{aligned}
x(k+1) &= Ax(k) + B\kappa(\hat{x}(k), \hat{d}(k), r) + B_d d \\
\hat{x}(k+1) &= (A + L_x C)\hat{x}(k) + (B_d + L_x C_d)\hat{d}(k) \\
&\quad + B\kappa(\hat{x}(k), \hat{d}(k), r) - L_x y(k) \\
\hat{d}(k+1) &= L_d C\hat{x}(k) + (I + L_d C_d)\hat{d}(k) - L_d y(k)
\end{aligned}
$$

converges, i.e. $\hat{x}(k) \to \hat{x}_\infty$, $\hat{d}(k) \to \hat{d}_\infty$, $y(k) \to y_\infty$ as $k \to \infty$.

Then $z(k) = Hy(k) \to r$ as $k \to \infty$.

[U. Mäder, F. Borrelli, M. Morari, Offset-free Model Predictive Control, Automatica 2009]

Offset-free control: Example

Double Integrator: first set point: $r = 0$, second set point: $r = 3$
1 output, 1 input, 1 modeled disturbance

Standard tracking:

Model:

$$x(k+1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x(k) + \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} u(k)$$

$$z(k) = y(k) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(k)$$

$$-0.5 \leq u(k) \leq 0.5$$

$$-5 \leq y(k) \leq 5$$

Target condition:

$$\begin{bmatrix} I - A & -B \\ C & 0 \end{bmatrix} \begin{bmatrix} x_s \\ u_s \end{bmatrix} = \begin{bmatrix} 0 \\ r \end{bmatrix}$$

Offset-free tracking:

Model:

$$x(k+1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x(k) + \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} u(k) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} d(k)$$

$$z(k) = y(k) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(k)$$

$$-0.5 \leq u(k) \leq 0.5$$

$$-5 \leq y(k) \leq 5$$

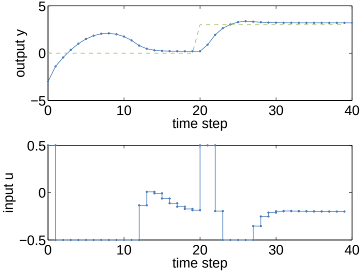
$\begin{bmatrix} I - A & -B_d \\ C & 0 \end{bmatrix}$ has full column rank

Target condition:

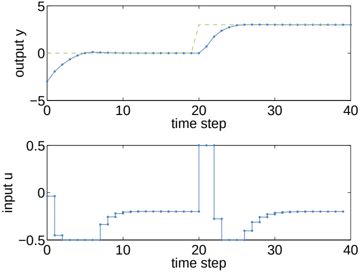
$$\begin{bmatrix} I - A & -B \\ C & 0 \end{bmatrix} \begin{bmatrix} x_s \\ u_s \end{bmatrix} = \begin{bmatrix} B_d \hat{d} \\ r - C_d \hat{d} \end{bmatrix}$$

Offset-free control: Example

Standard tracking



Offset-free tracking



Summary: Offset-free control

Approach for offset-free constrained control:

- Model the disturbance e.g. using integrating disturbance dynamics
- Use the output measurements and model to estimate the state and the disturbance (e.g. Linear estimator, Kalman filter)
- Find control inputs that use the disturbance estimate to remove offset:
  - Modify system steady state to account for effect of disturbance
  - Modify target to account for effect of disturbance on tracked variables
  - Solve MPC problem for regulation/tracking using the disturbance estimate

Main result: If

- number of disturbance states is equal to number of outputs,
- the target steady-state problem is feasible and no constants are active at steady-state
- the closed-loop system converges,

then the target is achieved without offset.

Outline

1. MPC: Practical Issues

2. Reference Tracking

3. Enlarging the Feasible Set

Outline

3. Enlarging the Feasible Set

   MPC without Terminal Set

   Soft Constrained MPC

Review: Stability of MPC

Assumptions

1. Stage cost is positive definite, i.e. it is strictly positive and only zero at the origin
2. Terminal set is **invariant** under the local control law $\kappa_f(x_i)$:

$$x_{i+1} = Ax_i + B\kappa_f(x_i) \in \mathcal{X}_f, \quad \text{for all } x_i \in \mathcal{X}_f$$

   All state and input **constraints are satisfied** in $\mathcal{X}_f$:

$$\mathcal{X}_f \subseteq \mathcal{X},\ \kappa_f(x_i) \in \mathcal{U}, \quad \text{for all } x_i \in \mathcal{X}_f$$

3. Terminal cost is a continuous **Lyapunov function** in the terminal set $\mathcal{X}_f$ and satisfies:

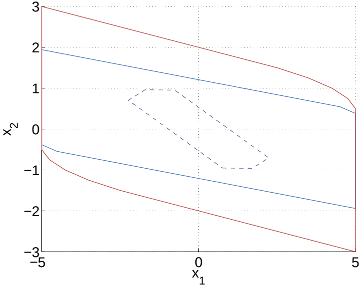
$$l_f(x_{i+1}) - l_f(x_i) \leq -l(x_i, \kappa_f(x_i)), \quad \text{for all } x_i \in \mathcal{X}_f$$

Thm: The closed-loop system under the MPC control law $u_0^\star(x)$ is stable and the system $x(k+1) = Ax(k) + Bu_0^\star(x(k))$ is invariant in the feasible set.

MPC without Terminal Set

Motivation

- Terminal constraint reduces feasible set



- Blue line: Feasible set with terminal constraint
- Red line: Feasible set without terminal constraint
- Dashed line: Terminal set

- Potentially adds large number of extra constraints
- Adds state constraints to problems with only input constraints

Goal: MPC without terminal constraint with guaranteed stability

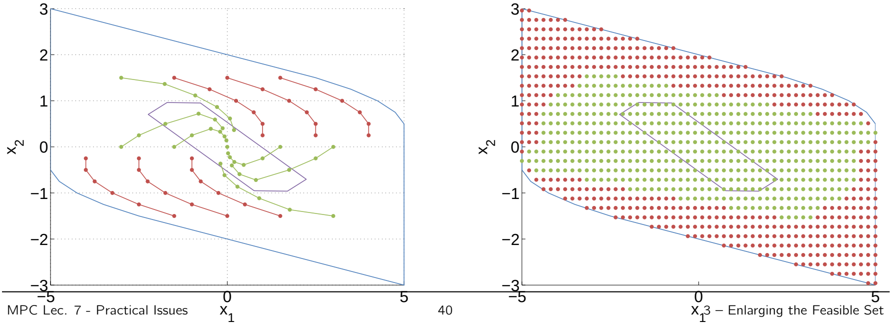
Note: Feasible set without terminal constraint is not invariant.

MPC without Terminal Set

We can remove terminal constraint while maintaining stability if

- initial state lies in sufficiently small subset of feasible set
- N is sufficiently large

such that terminal state satisfies terminal constraint without enforcing it in the optimization.
⇒ Solution of the finite horizon MPC problem corresponds to the infinite horizon solution



MPC without Terminal Set: Discussion

Advantage: Controller defined in a larger feasible set

Disadvantage: Characterization of region of attraction or specification of required horizon length extremely difficult

Remarks:

- Terminal constraint provides a sufficient condition for stability:
  Region of attraction without terminal constraint may be larger than for MPC with terminal constraint
- In practice: Enlarge horizon and check stability by sampling
- With larger horizon length N, region of attraction approaches maximum control invariant set

Outline

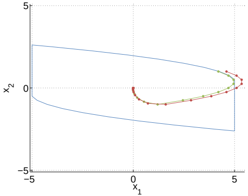
3. Enlarging the Feasible Set

    MPC without Terminal Set

    Soft Constrained MPC

Soft Constraints: Motivation

- Input constraints are dictated by physical constraints on the actuators and are usually "hard"
- State/output constraints arise from practical restrictions on the allowed operating range and are **rarely hard**
- Hard state/output constraints always lead to **complications in the controller implementation**
  - Feasible operating regime is constrained even for stable systems
  - Controller patches must be implemented to generate reasonable control action when measured/estimated states move outside feasible range because of disturbances or noise



- In industrial implementations, typically, state constraints are **softened**

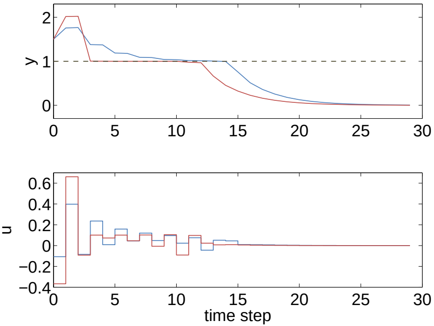
Objectives

Goals:

- Minimize the duration of the violation
- Minimize the size of the violation

These can be conflicting goals, i.e. reduction in size of violation can only be achieved at cost of large increase in duration of violation

$\rightarrow$ Multi-objective problem

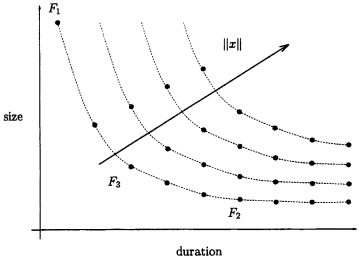


Red: Minimum time of violation

Blue: Minimum size of violation

Multi-objective Problem

For given system and horizon can plot pareto optimal size/duration curve for different initial conditions:



Best operation points lie on pareto optimal curve:

- points below cannot be attained
- points above are inferior

$\rightarrow$ Operation at pareto optimality is in general difficult and only approximately achieved

Best operation point minimum time or minimum violation?

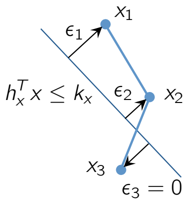
Depends on application, e.g.:

- if product must be discarded during constraint violation, goal is minimum time of violation
- if large constraint violations can lead to process shutdown or exceptions goal is minimum amount of violation

Soft Constrained MPC Problem Setup

$$\min_{\mathbf{u}} \sum_{i=0}^{N-1} x_i^T Q x_i + u_i^T R u_i + l_\epsilon(\epsilon_i) + x_N^T P x_N + l_\epsilon(\epsilon_N)$$

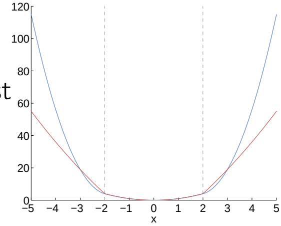
$$\begin{aligned}
\text{s.t. } & x_{i+1} = Ax_i + Bu_i \\
& H_x x_i \le k_x + \epsilon_i, \\
& H_u u_i \le k_u, \\
& \epsilon_i \ge 0
\end{aligned}$$



- Relax state constraints by introducing so called **slack variables** $\epsilon_i \in \mathbb{R}^p$
- Penalize amount of constraint violation in the cost by means of penalty $l_\epsilon(\epsilon_i)$

How to choose penalty?

- Quadratic penalty: $l_\epsilon(\epsilon_i) = \epsilon_i^\top S \epsilon_i$
- Quadratic and linear norm penalty: $l_\epsilon(\epsilon_i) = \epsilon_i^\top S \epsilon_i + v\|\epsilon_i\|_{1/\infty}$



Blue: $x^T x + 10\epsilon^T \epsilon$
Red: $x^T x + 10\|\epsilon\|_1$

Mathematical Formulation

- **Original** problem:

$$\min_{z} \; f(z)$$
$$\text{subj. to } g(z) \leq 0$$

  Assume for now $g(z)$ is scalar valued.

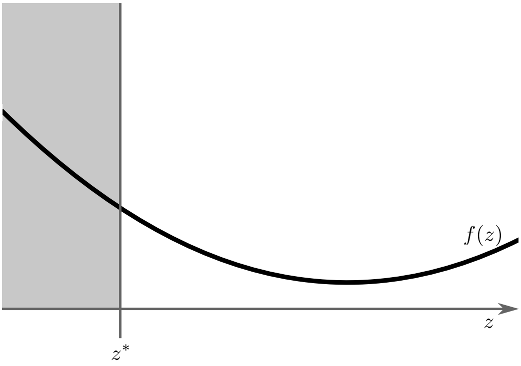
- **"Softened"** problem:

$$\min_{z,\epsilon} \; f(z) + l_\epsilon(\epsilon)$$
$$\text{subj. to } g(z) \leq \epsilon$$
$$\epsilon \geq 0$$

Requirement on $l_\epsilon(\epsilon)$
If the original problem has a feasible solution $z^\star$, then the softened problem should have the same solution $z^\star$, and $\epsilon = 0$.

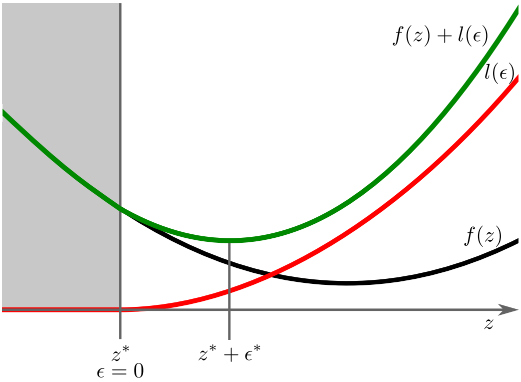
Note: $l_\epsilon(\epsilon) = s \cdot \epsilon^2$ does not meet this requirement for any $s > 0$ as demonstrated next.

Quadratic Penalty



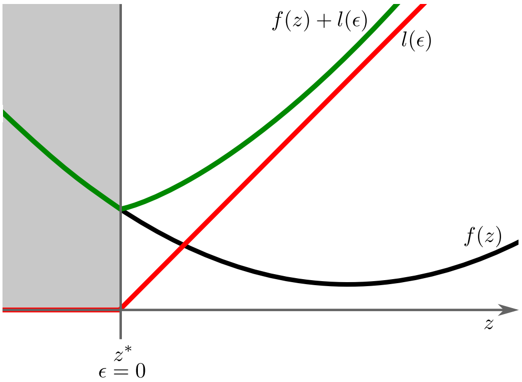
- Constraint function $g(z) \triangleq z - z^\star \leq 0$ induces feasible region (grey) $\implies$ minimizer of the original problem is $z^\star$

Quadratic Penalty



- Constraint function $g(z) \triangleq z - z^\star \leq 0$ induces feasible region (grey)
  $\Longrightarrow$ minimizer of the original problem is $z^\star$
- **Quadratic penalty** $l_\epsilon(\epsilon) = s \cdot \epsilon^2$ for $\epsilon \geq 0$, $l_\epsilon(\epsilon) = 0$ otherwise
  $\Longrightarrow$ minimizer of $f(z) + l_\epsilon(\epsilon)$ is $(z^\star + \epsilon^\star, \epsilon^\star)$ instead of $(z^\star, 0)$

Linear Penalty



- Constraint function $g(z) := z - z^\star \leq 0$ induces feasible region (grey)
  $\Longrightarrow$ minimizer of the original problem is $z^\star$
- **Linear penalty** $l_\epsilon(\epsilon) = v \cdot \epsilon$ for $\epsilon \geq 0$, $l_\epsilon(\epsilon) = 0$ otherwise,
  with v chosen large enough so that $v + \lim_{z \to z^\star} f'(z) > 0$
  $\Longrightarrow$ minimizer of $f(z) + l_\epsilon(\epsilon)$ is $(z^\star, 0)$

Main Result

Theorem: Exact Penalty Function

$l_\epsilon(\epsilon) = v \cdot \epsilon$ satisfies the requirement for any $v > \lambda^\star \geq 0$, where $\lambda^\star$ is the optimal Lagrange multiplier for the original problem.

- In practice, combined cost is used for exact penalty and tuning capabilities

$$l_\epsilon(\epsilon) = v \cdot \epsilon + s \cdot \epsilon^2$$

with $v > \lambda^\star$ and $s > 0$.

- Extension to multiple constraints $g_j(z) \leq 0,\ j = 1, \ldots, r$:

$$l_\epsilon(\epsilon) = v \cdot \|\epsilon\|_{1/\infty} + \epsilon^T S \epsilon \tag{1}$$

where $\epsilon = [\epsilon_1, \ldots, \epsilon_r]^T$ and $v > \|\lambda^\star\|_D$ and $S \succ 0$ can be used to weight violations differently. $\|\cdot\|_D$ denotes the dual of the chosen norm (the dual of $\|\cdot\|_1$ is $\|\cdot\|_\infty$ and vice versa). [2]

[2] Further reading: Soft Constraints and Exact Penalty Functions in Model Predictive Control, UKACC International Conference (Control), 2000

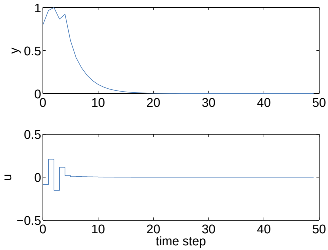
Example

Consider the third-order non-minimum phase system

$$x(k+1) = \begin{bmatrix} 2 & -1.45 & 0.35 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} x(k) + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u(k)$$
$$y(k) = \begin{bmatrix} -1 & 0 & 2 \end{bmatrix} x(k)$$

- Constraint: $-1 \leq y \leq 1$
- Controller parameters: $Q = C^\top C$, $R = 1$ and $N = 20$.

Initial condition $x(0) = \begin{bmatrix} 0.8 & 0.8 & 0.8 \end{bmatrix}^\top$:

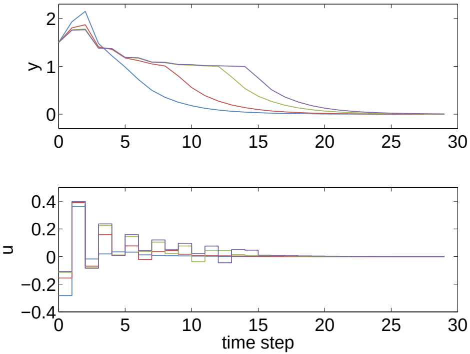


Soft constraints with quadratic penalty

Properties of the quadratic penalty:

- Well-posed quadratic program (positive definite Hessian)
- Increase in S leads to 'hardening' of the soft constraints

Example:



Initial condition:
$x(0) = \begin{bmatrix} 1.5 & 1.5 & 1.5 \end{bmatrix}^\top$

Blue:	$S = 1$
Red:	$S = 10$
Green:	$S = 50$
Violet:	$S = 100$

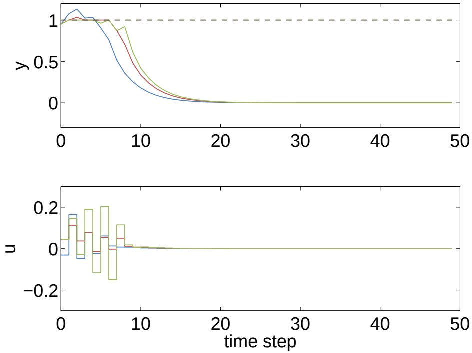
$\rightarrow$ Increase in S leads to reduced size of violation but longer duration

Soft constraints with quadratic and linear penalty

Properties of the linear penalty:

- Allows for exact penalties: If weight v is chosen large enough, constraints are satisfied if possible

Example:



Initial condition:
$x(0) = \begin{bmatrix} .95 & .95 & .95 \end{bmatrix}^\top$
Penalty: 1-norm, S= 20

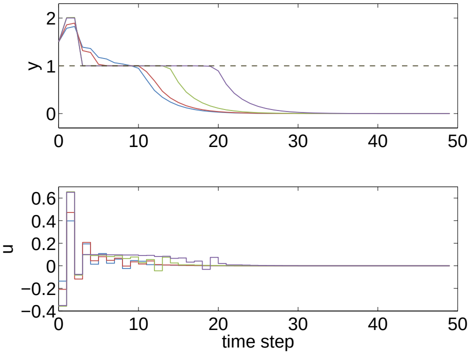
Blue:	$v = 0.1$
Red:	$v = 10$
Green:	$v = 25$

Soft constraints with quadratic and linear penalty

Properties of the linear penalty:

- Allows for exact penalties: If weight v is chosen large enough, constraints are satisfied if possible
- Increasing v results in increasing peak violations and decreasing duration
- Large linear penalties make tuning difficult and cause numerical problems

Example:



Initial condition:
$x(0) = \begin{bmatrix} 1.5 & 1.5 & 1.5 \end{bmatrix}^\top$
Penalty: 1-norm, S= 20

Blue: $v = 0.1$
Red: $v = 10$
Green: $v = 25$

Simplification: Separation of Objectives

1. Minimize violation over horizon:

$$
\begin{aligned}
\boldsymbol{\epsilon}^{\min} = \underset{\mathbf{u},\boldsymbol{\epsilon}}{\operatorname{argmin}} \sum_{i=0}^{N-1} \epsilon_i^T S \epsilon_i + v^\top \epsilon_i \\
\text{s.t. } & x_{i+1} = A x_i + B u_i \\
& H_x x_i \le k_x + \epsilon_i, \\
& H_u u_i \le k_u, \\
& \epsilon_i \ge 0
\end{aligned}
$$

2. Optimize controller performance:

$$
\begin{aligned}
\min_{\mathbf{u}} \sum_{i=0}^{N-1} x_i^T Q x_i + u_i^T R u_i + x_N^T P x_N \\
\text{s.t. } & x_{i+1} = A x_i + B u_i \\
& H_x x_i \le k_x + \epsilon_i^{\min}, \\
& H_u u_i \le k_u,
\end{aligned}
$$

$\Rightarrow$ Advantage: Simplifies tuning, constraints will be satisfied if possible

$\Rightarrow$ Disadvantage: Requires solution of two optimization problems

Summary: Soft constraints

- Soft constraints recover feasibility of the optimization when constraints cannot be satisfied
- Allow for a variety of violation duration vs. size tradeoffs
- Good closed-loop properties

$\Rightarrow$ Generally applied in practice

Note: standard methods for soft constrained MPC do not provide a stability guarantee for open-loop unstable systems

$\Rightarrow$ Developments towards soft constrained MPC with stability guarantees[3][4]

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[3] K. P. Wabersich, R. Krishnadas and M. N. Zeilinger, A Soft Constrained MPC Formulation Enabling Learning From Trajectories With Constraint Violations, IEEE Control Systems Letters, vol. 6, pp. 980-985, 2022

[4] M.N. Zeilinger, M. Morari and C.N. Jones, Soft constrained model predictive control with robust stability guarantees, IEEE Transactions on Automatic Control, 2014

Putting it all together

- In general state cannot be measured:
  - Use Kalman filter to estimate the state
- Design tracking problem:
  - Rewrite problem in delta-formulation
  - Setup target steady-state problem
  - Calculate terminal weight and scale terminal constraint to guarantee convergence
- Extend to offset-free tracking:
  - Augment model including a disturbance model
  - Augment the estimator to estimate the state and the disturbance
  - Adapt target steady-state problem using the disturbance estimate
- Possibly: Remove terminal constraint while choosing long horizon
- Introduce soft constraints to ensure feasibility at all times:
  - Introduce slack variables for constraint relaxation
  - Choose penalty on slack variables (quadratic, linear)